

IV-1: A Two-Stage Strake-Stabilised Interceptor

Coupled nTop and SUAVE conceptual sizing, and three requirements that could not hold together

Prepared with nTop Automate 5.53.2 / 5.54.0 and SUAVE 2.5.2. All geometry is authored programmatically. Report generated from the run artefacts under runs/IV-1.

Summary

This report describes a conceptual sizing loop for a two-stage, strake-stabilised interceptor-class vehicle. SUAVE does the physics. nTop does the geometry. The two are coupled: nTop measures the solid it builds, and those measurements go back into the mass and aerodynamic models.

The loop found a design that meets all 15 recorded constraints. It weighs 592.2 kg. It reaches 160.9 km of slant range, which is 100.0 statute miles. It arrives at 19.9 km and Mach 4.00.

The result matters less than what the loop caught. Three requirements were mutually exclusive, and the cause was an exclusion in the specification itself. One structural limit was unachievable anywhere in the design space. One modelling assumption about the propellant grain turned out to be the binding physical limitation on the whole vehicle. The constraint list had a hole. The atmosphere table stopped below the altitude the vehicle flies at. Section 1 gives each finding.

The vehicle and its requirements are invented for this demonstration. They correspond to no real programme. This is a parametric configuration study built from published conceptual-design methods. It is not a model of any fielded system. Section 6 lists every limitation and every value that is a guess.

Quantity	Value	Requirement	Status
Slant range at intercept	160.9 km (100.0 mi)	≥ 160.9 km	met, active
Intercept altitude	19.9 km	≥ 15.0 km	met
Mach at intercept	4.00	≥ 3.00	met
Lateral acceleration available	15.86 g	≥ 15.0 g	met, by divert motor
Launch mass	592.2 kg	≤ 1400 kg	met
Peak dynamic pressure	304.6 kPa	≤ 350 kPa	met, limit revised
Stacked length	5.28 m	≤ 5.40 m	met
Maximum body diameter	0.42 m	≤ 0.42 m	met, active
Mass jettisoned at separation	31.3 kg	not constrained	-

Table 1. Headline result. The geometry is measured by nTop inside the loop. Slant range and maximum diameter both sit exactly on their limits. Source: runs/IV-1/converged.json.

1. What the loop caught

A sizing loop earns its cost when it finds errors that inspection misses. This one found five. Each was found by walking trajectories against the constraint set, not by reading the specification.

1.1 Three requirements were mutually exclusive

As first written, A2, A3 and A11 could not hold together. A2 asks for 160.9 km of slant range, which is 100 statute miles. A3 asks for an intercept at or above 15 km. A11 asks for 15 g of lateral acceleration at that intercept.

The specification assumed purely aerodynamic control. Available lateral acceleration is then q times S_{ref} times CN_{max} , divided by mass. That needs dynamic pressure. Dynamic pressure needs air. A2 needs a lofted trajectory to reach that far, and lofting puts the intercept where there is no air. The two pull in opposite directions.

The audit swept the pitchover angle on the stack that closes every motor, volume and structural constraint: 732 kg, 5.28 m, 1,404 kN.s of vacuum impulse. It then walked each trajectory for the furthest point at which A3, A4 and A11 all hold at once.

Pitchover [deg]	Furthest point meeting A3, A4 and A11	Altitude there	Max slant range	Altitude at max slant
12	none	-	80.7 km	0.0 km
16	none	-	102.4 km	0.0 km
20	58.3 km	18.5 km	140.0 km	0.0 km
24	52.6 km	20.9 km	160.9 km	21.9 km
28	47.7 km	22.3 km	160.9 km	41.6 km
32	43.8 km	23.4 km	160.9 km	56.5 km
40	32.2 km	21.3 km	160.9 km	81.4 km
50	23.3 km	18.4 km	160.9 km	107.7 km

Table 2. Pitchover sweep. The lateral-acceleration figures behind this table use a generous CN_{max} of 2.5 rather than the build-up, so every entry is an upper bound on what the vehicle can do.

Source: SPEC_IV1.md section 2, reproduced by `scripts/iv1_envelope_probe.py`.

The best slant range meeting every requirement is 58.3 km, which is 36.2 miles against the 100 miles A2 asks for. The shortfall is 102.7 km, a factor of 2.76.

The cause is not the design vector. It is a hard physical ceiling. At the post-separation mass of 258 kg, and with the same generous CN_{max} of 2.5, 15 g is available only below these altitudes:

Mach	15 g available only below
3.0	10.0 km
4.0	13.7 km
5.0	16.6 km

Table 3. The altitude ceiling A11 imposes on its own, independent of any trajectory. A3 requires the intercept at or above 15 km, so the band in which A3 and A11 overlap is empty at Mach 3 to 4.

Stack: 732 kg, 5.28 m, 1404 kN.s of vacuum impulse, 47 kg jettisoned. Every lateral-acceleration figure here is an UPPER bound, because the audit uses a generous CN_{max} of 2.5 in place of the build-up. Source: SPEC_IV1.md section 2, reproduced in runs/IV-1/figures/evidence_iv1.json.

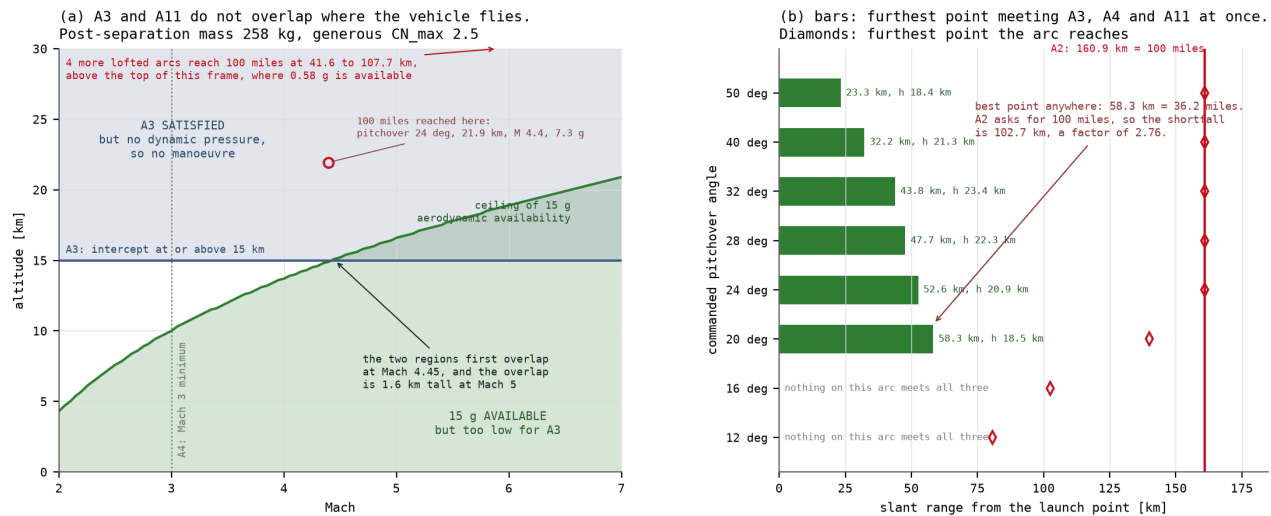


Figure 1. The conflict, drawn. Panel (a) puts both requirements in the same axes: the green region is where 15 g is available, the blue region is where A3 puts the intercept, and they first touch at Mach 4.45. Every trajectory that reaches 100 miles does so far above the green region. Panel (b) shows what that costs in range.

Resolution: the vehicle needs lateral control that does not depend on dynamic pressure. That is a divert or attitude-control motor, which is what vehicles of this class carry and precisely why they carry it. **SPEC_IV1.md section 8 originally excluded attitude-control thrusters, and that exclusion is what made the requirement set infeasible.** A11 was therefore restated on the capability rather than on the mechanism, and A13 was added so the motor is sized, reported and charged to the mass statement.

The converged vehicle carries a divert motor of 39.0 kN for 3.0 s, which is 117 kN.s of total impulse and 78.8 kg of mass. At the intercept it supplies 15.86 g. The airframe supplies only 7.45 g there, because the dynamic pressure has fallen to 63 kPa.

The conclusion is a design conclusion, not a design. A purely aerodynamic interceptor at this size cannot engage at 100 miles and still manoeuvre. That statement is more useful than the vehicle that follows from it.

1.2 The dynamic-pressure limit was unachievable

A10 was first written as 250 kPa. Nothing in the design space meets it. A sweep over stage propellant, booster thrust, pitchover angle and pitchover time found a floor of about 278 kPa on peak dynamic pressure, rising to 309 kPa for any design that also meets A2, A3 and A4.

The reason is the launch condition. A1 launches vertically from a canister at sea level, so the vehicle accelerates through dense air near the ground. The converged design reaches its peak of 304.6 kPa at 4.61 km and Mach 2.77, only 12.6 s after launch. Holding 250 kPa would cap the vehicle near Mach 2.75 at 6 km, which cannot reach 100 miles.

A10 was therefore revised to 350 kPa, about 13 percent above the 309 kPa floor that A2, A3 and A4 impose. **The consequence must be stated plainly. The structural model in this toolkit is wall thickness times density, plus a hoop-stress check on the motor case and a buckling check on the interstage. It does not size the**

airframe for a 350 kPa aerodynamic load. The limit is a stated requirement that the mass model does not verify, so the airframe mass is optimistic by an amount this toolkit cannot quantify.

1.3 The tubular grain is the binding physical limitation

Every stage grain in this model is an internal-burning tube, closed at its mean web. A tube is the least area-efficient internal-burning geometry. Real tactical motors of this thrust class use slotted, star or finocyl grains, which reach several times the burning area in the same case length. **No shape factor is applied here, because no sourced multiplier was available.** The model reports the long tube instead, so the assumption surfaces as a design constraint rather than as silent optimism.

That assumption then drove three separate decisions in the design.

Booster thrust [kN]	Grain L/D	Grain length [m]	Bay length [m]	Nozzle exit [m]	Grain closes
45	2.61	1.063	2.50	0.164	yes
60	3.10	1.263	2.50	0.189	yes
80	3.82	1.558	2.50	0.219	yes
100	4.57	1.863	2.50	0.245	yes
110	4.95	2.017	2.50	0.257	yes
120	5.33	2.173	2.50	0.268	yes
150	6.48	2.642	2.50	0.300	NO
200	8.41	3.429	2.50	0.346	NO

Table 4. Booster-thrust sweep on the converged geometry. A tubular grain needs burning area, and burning area on a tube needs length. The grain runs out of bay between 120 and 150 kN. The converged design uses 45 kN.

Stage-2 propellant [kg]	Volumetric loading	Web [mm]	Bay radius [mm]	Grain closes
70	0.694	120	164	yes
80	0.793	138	164	yes
90	0.892	155	164	yes
95	0.942	163	164	yes
100	0.991	172	164	NO
110	1.090	189	164	NO
130	1.289	224	164	NO

Table 5. Stage-2 propellant sweep on the converged geometry. Above 95 kg the web is wider than the bay radius, so the tube cannot hold the charge at all. The converged design carries 90 kg.

The nozzle is the third consequence. The starting design vector used an area ratio of 10 on a 0.40 m booster at 170 kN. That gives an exit diameter of 409 mm, which is wider than the 400 mm body. The exit cone does not fit inside the airframe. The booster was therefore taken to the full 0.42 m allowed diameter and its area ratio was reduced to 6.

The same starting vector also needed 151.1 percent volumetric loading on stage 2. The model reports all three problems as warnings rather than smoothing them over, and a test asserts that it does.

1.4 The constraint list had a hole

The constraint list gated the grain length-to-diameter ratio but not grain closure. Those are different questions. A grain can sit inside the L/D band of 1.0 to 8.0 and still be impossible, because a tubular grain needs a bore for burning area and the remaining web has to fit inside the bay radius.

Gating only on L/D let a stage-2 grain through at 134 percent volumetric loading, with a 232 mm web in a 164 mm bay radius. Two constraints were added per stage: volumetric loading at or below 1.0, and a closure flag from the grain solver. The converged design now sits at 0.892 on stage 2, which is the third tightest constraint in the whole set.

A second hole is still open, and this report does not close it. Requirement A9 asks for a static margin of at least 1.0 calibre in each stage over the whole flight. A9 is not among the 15 constraints the sizing script records. Section 2.2 evaluates it here and reports the result. It does not pass.

1.5 The atmosphere table stopped below the flight path

The cached US Standard 1976 table stopped at 30 km and clamped above it. That was ample for the SV-1 cruise-and-dive mission. It is not ample for a lofted two-stage intercept. Measured IV-1 arcs apogee between 45 and 54 km, and the converged design apogees at 37.9 km.

A clamp holds density at its ceiling value, so it overstates drag on everything above the ceiling. At 37.9 km the clamp overstated density by a factor of 3.4. At 50 km it overstated it by a factor of 17.9, which is more than an order of magnitude.

The table now runs to 86 km, the upper limit of the standard's lower atmosphere, on a 10 m grid of 8,601 nodes. The converged trajectory now flies entirely inside the table: the recorded overshoot above the ceiling is 0.0 m.

An instructive near-miss

A first comparison against the published tables showed SUAVE about 4.4 percent off at 47 km. That looked like a model defect. It was not. It was an error in the comparison.

Step	Value
Pressure SUAVE returns at 47 km geometric	115.84 Pa
Published 47 km row, which is geopotential	110.906 Pa
Apparent error of the naive comparison	+4.45 %
Geopotential altitude of 47 km geometric	46.655 km
Pressure at the geometric altitude for 47 km geopotential	110.895 Pa

Table 6. The geopotential-against-geometric near-miss. The standard is tabulated against geopotential altitude. SUAVE takes geometric altitude and converts internally. Doing the conversion properly removes the whole discrepancy. A test now asserts that the naive comparison still fails, so nobody can fix a defect that does not exist.

2. The converged design

IV-1 stacked assembly, 5.08 m: ogive payload stage with four strakes and four tail fins, conical interstage, booster with four tail fins

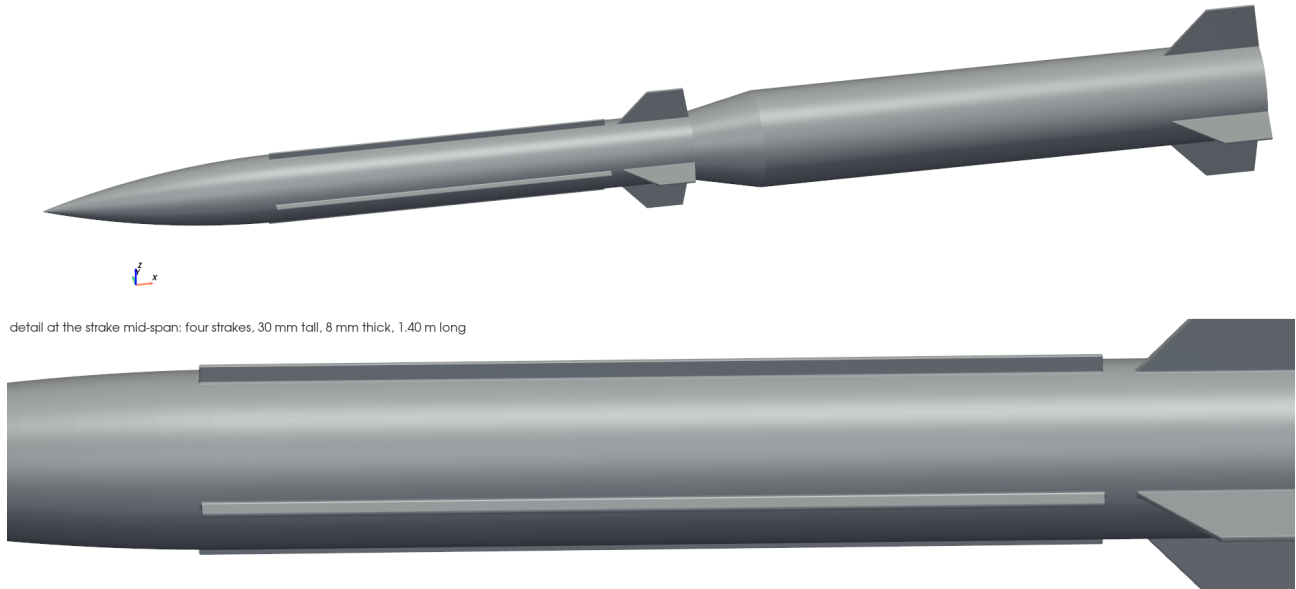


Figure 2. The IV-1 stack, rendered from the STL that nTop exported. Upper panel: ogive payload stage with four strakes and four tail fins, conical interstage, booster with four tail fins. Lower panel: the strake mid-span, where the 30 mm strake and its root junction into the wall are visible. This render is of the starting design vector at 5.08 m, not of the converged 5.28 m stack; the topology is the same and the dimensions differ. No one opened the nTop graphical interface.

Parameter	Stage 1, booster	Stage 2, payload stage	Unit
Body diameter	0.420	0.340	m
Stage length	2.60	2.40	m
Propellant mass	220.0	90.0	kg
Thrust	45.0	18.0	kN
Chamber pressure	8.0	8.0	MPa
Nozzle area ratio	6.0	18.0	-
Wall thickness	3.20	2.60	mm
Fin count	4	4	-
Fin exposed semi-span	0.200	0.140	m
Fin root chord	0.400	0.300	m
Fin leading-edge sweep	42.0	42.0	deg
Jettisoned at separation	yes	no	-

Table 7. The converged design vector, by stage.

Parameter	Value	Unit
Stacked length	5.28	m
Maximum body diameter	0.42	m
Payload-stage nose fineness	3.00	-
Payload-stage nose length	1.020	m
Interstage length	0.280	m
Interstage wall thickness	2.50	mm
Strake count	4	-
Strake height above the body	30.0	mm
Strake length	1.40	m
Strake thickness	8.0	mm
Strake leading-edge station	0.95	m
Commanded pitchover angle	38.0	deg
Pitchover start time	5.0	s
Commanded pitch rate	8.0	deg/s
Divert-motor thrust	39.0	kN
Divert-motor firing time	3.0	s
Divert-motor total impulse	117.1	kN.s

Table 8. Stack-level parameters, strakes and the ascent programme.

2.1 The motor stack

Quantity	Stage 1	Stage 2	Unit
Vacuum thrust	47.1	18.0	kN
Sea-level thrust	45.0	15.7	kN
Vacuum specific impulse	275.0	297.5	s
Burn time	12.58	14.59	s
Mass flow	17.48	6.17	kg/s
Throat diameter	67.0	39.8	mm
Exit diameter	164.1	168.8	mm
Grain length	1.063	0.591	m
Grain length over diameter	2.61	1.80	-
Grain web	134	155	mm
Volumetric loading	0.375	0.892	-
Total impulse, vacuum	593	263	kN.s

Table 9. Motor operating points. Stage-1 thrust is a sea-level equivalent, because A1 launches the stack from a canister at sea level. Stage-2 thrust is a vacuum value, because it lights above 4 km. Each stage runs one throat area for its whole burn, so the stack needs no throat-changing mechanism at all. That is the opposite of the SV-1 dual-thrust motor, which needs a throat that shrinks and cannot have one.

2.2 Mass statement

The launch mass is 592.2 kg. The booster group weighs 292.8 kg and the interstage 3.02 kg. At separation 31.3 kg leaves the vehicle, and the payload stage continues at 217.6 kg.

Stage	Item	Mass [kg]	Percent	Station [m]	Provenance
2	Propellant	90.00	15.2	2.070	analytic
2	Payload	75.00	12.7	1.540	requirement
2	Airframe structure and surfaces	24.17	4.1	1.518	ntop measured
2	Radome	11.52	1.9	0.459	analytic
2	Dry-mass contingency	8.03	1.4	1.360	correlation
2	Motor insulation	3.56	0.6	2.070	analytic
2	Motor case	1.87	0.3	2.070	analytic
2	Nozzle assembly	1.76	0.3	2.280	correlation
2	Motor hardware not modelled	1.25	0.2	2.070	correlation
2	Igniter	0.45	0.1	1.806	analytic
1	Propellant	220.00	37.1	4.084	analytic

Stage	Item	Mass [kg]	Percent	Station [m]	Provenance
1	Airframe structure and surfaces	39.96	6.7	4.171	ntop measured
1	Dry-mass contingency	11.11	1.9	4.149	correlation
1	Motor hardware not modelled	7.86	1.3	4.084	correlation
1	Motor insulation	6.76	1.1	4.084	analytic
1	Motor case	4.56	0.8	4.084	analytic
1	Nozzle assembly	1.47	0.2	5.150	correlation
1	Igniter	1.10	0.2	3.127	analytic
-	Interstage	3.02	0.5	2.540	analytic
2	Attitude-control motor pack	78.78	13.3	not stationed	correlation
	TOTAL	592.24	100.0		

Table 10. Group-weight statement, by stage. Provenance says where each mass came from. Only 12.5 percent of the launch mass is measured by nTop, which is 64.13 kg. The propellant, the payload and the motor correlations are not geometry, so they cannot be measured. The divert-motor pack is charged on top of the stage statement and carries no station, so it is listed separately.

Launch mass 592.2 kg. nTop-measured share 12.5 percent (64.13 kg). Jettisoned at separation 31.3 kg, so the payload stage weighs 217.6 kg after separation. The propellant, the payload and the motor correlations do not come from geometry, which is why the measured share is small. Source: runs/IV-1/converged.json.

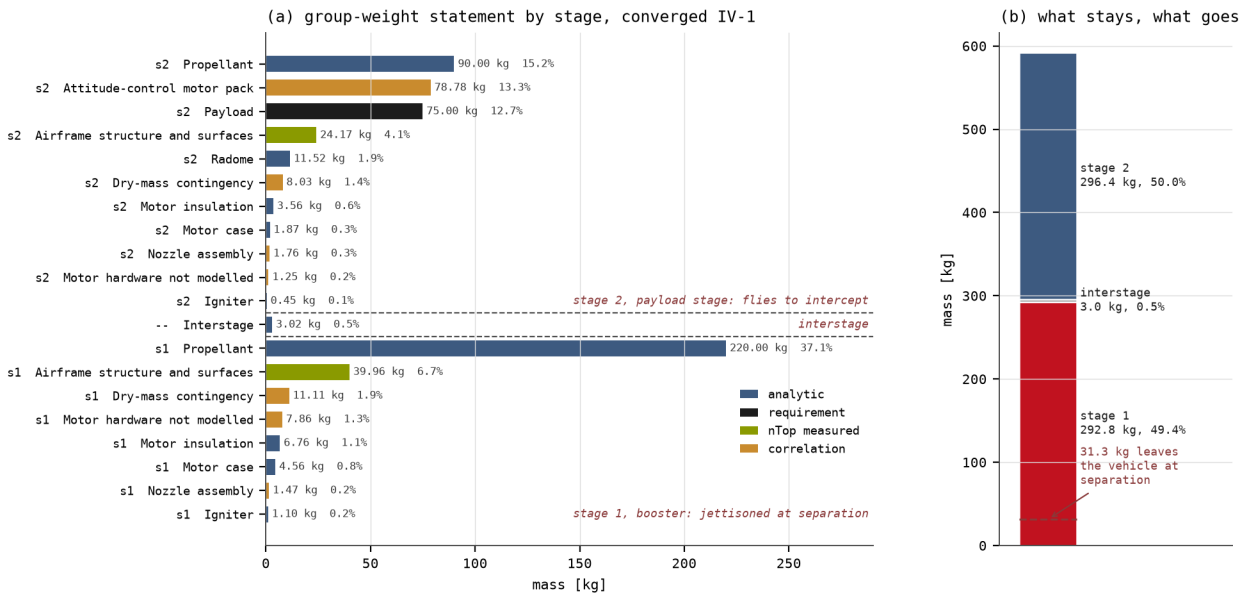


Figure 3. The mass statement, grouped by stage and coloured by provenance. Panel (b) is the part that a single-stage vehicle does not have: half the launch mass is booster, and 31.3 kg of it leaves the vehicle at separation.

2.3 Constraints

Constraint	Value	Sense	Limit	Margin [%]	Status
A6 max diameter [m]	0.420	<=	0.420	+0.0	met
stage 1 grain closes	1.000	>=	1.000	+0.0	met
stage 2 grain closes	1.000	>=	1.000	+0.0	met
A2 slant range [m]	160,934.000	>=	160,934.000	+0.0	met
A7 stacked length [m]	5.280	<=	5.400	+2.2	met
A11 lateral g	15.865	>=	15.000	+5.8	met
stage 2 vol loading	0.892	<=	1.000	+10.8	met
A10 q_max [Pa]	304,577.260	<=	350,000.000	+13.0	met
A3 intercept alt [m]	19,948.098	>=	15,000.000	+33.0	met
A4 intercept Mach	4.004	>=	3.000	+33.5	met
A8 launch mass [kg]	592.238	<=	1,400.000	+57.7	met
stage 1 vol loading	0.375	<=	1.000	+62.5	met
grain L/D stage 1	2.608	<=	8.000	+67.4	met
A12 s1 burnout alt [m]	4,664.224	<=	20,000.000	+76.7	met
grain L/D stage 2	1.797	<=	8.000	+77.5	met

Table 11. All 15 recorded constraints, tightest first, with margin as a percentage of each limit. Slant range and maximum diameter sit exactly on their limits by construction: the run terminates on the slant range, and the booster was taken to the full allowed diameter so its nozzle exit would fit.

* A9 is NOT one of the fifteen constraints the sizing script records. The value shown is the worst static margin computed for this report over four flight configurations and three Mach numbers, about the mass-statement centre of gravity. It is an open item, not a verdict: see section 6. Source: runs/IV-1/converged.json.

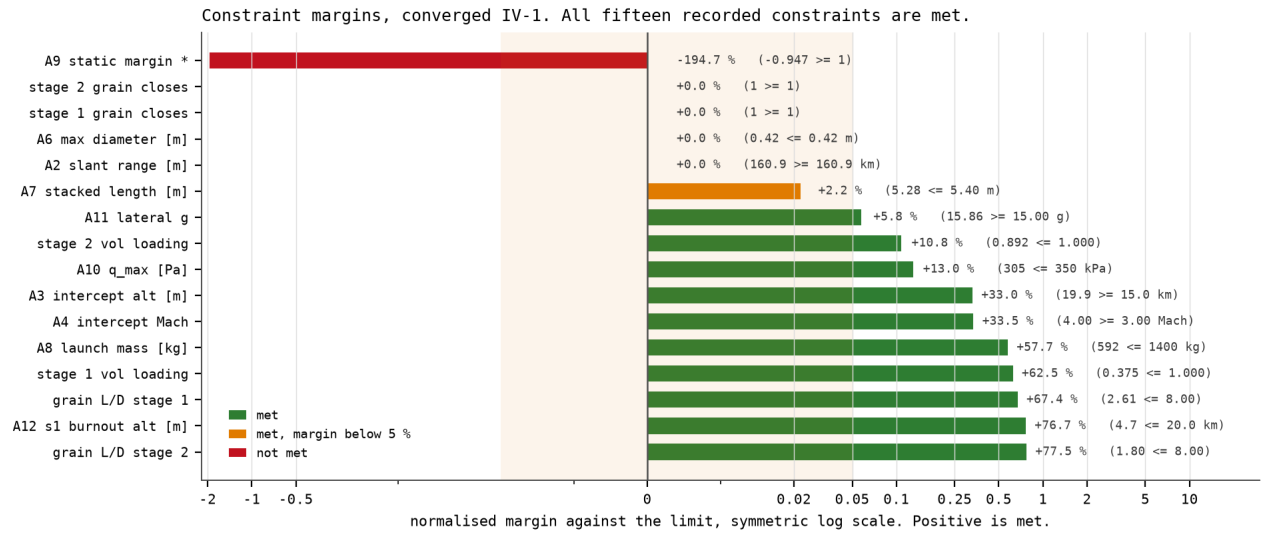


Figure 4. Constraint margins for the converged design, on a symmetric log scale. The three constraints at zero margin are met exactly. A9 is drawn in red because it is not one of the recorded constraints and does not pass; see the next paragraph.

Requirement A9 is evaluated here, and it fails. A9 asks for a static margin of at least 1.0 calibre in each stage over the whole flight. The sizing script does not record it as a constraint. This report computes it from the aerodynamic centre of pressure at 10 degrees of angle of attack and the centre of gravity of the mass statement, in the two flight configurations. The worst value is -0.947 calibres, which is unstable.

Configuration	Stage	Mach	x_cp [m]	x_cg [m]	Static margin [cal]
stack at launch	1	1.5	3.107	3.080	0.064
stack at launch	1	3.0	2.970	3.080	-0.263
stack at launch	1	4.0	2.874	3.080	-0.492
stack at stage-1 burnout	1	1.5	3.107	2.328	1.855
stack at stage-1 burnout	1	3.0	2.970	2.328	1.528
stack at stage-1 burnout	1	4.0	2.874	2.328	1.300
payload stage after separation	2	1.5	1.467	1.716	-0.732
payload stage after separation	2	3.0	1.462	1.716	-0.747
payload stage after separation	2	4.0	1.394	1.716	-0.947
payload stage at intercept	2	1.5	1.467	1.466	0.003
payload stage at intercept	2	3.0	1.462	1.466	-0.012
payload stage at intercept	2	4.0	1.394	1.466	-0.213

Table 12. Static margin, computed for this report. Stations are measured aft from the payload-stage nose tip. Reference diameter is 0.42 m on the stack and 0.34 m on the payload stage. Two caveats: the divert-motor pack carries no station, so it is not in the centre of gravity, and the centre of pressure is a linear build-up figure at one angle of attack. The result is an open item, not a

verdict. It is reported because a constraint that nobody evaluates is worse than one that fails.

2.4 The ascent

The mission has four phases: a vertical rise, a bounded-rate pitchover, staged boost, and a lofted midcourse coast. The flight lasts 157.8 s. It apogees at 37.9 km and reaches the slant-range condition on the way down, at 19.9 km and Mach 4.00. Reaching the range while descending is a legitimate intercept for this vehicle class, so the run does not terminate on ground impact first.

Event	Time [s]	Altitude [km]	Mach	Mass before [kg]	Mass after [kg]
pitchover	5.00	0.86	1.050	505.26	505.26
pitchover complete	11.76	4.16	2.524	387.16	387.16
stage 1 burnout	12.71	4.66	2.766	372.23	372.23
separation	13.31	4.98	2.708	372.23	340.97
stage 2 ignition	13.31	4.98	2.708	340.97	340.97
stage 2 burnout	28.02	14.14	4.862	250.97	250.97
apogee	96.74	37.94			
intercept	157.80	19.95	4.004		250.97

Table 13. The stage events. Separation and stage-2 ignition are the same instant, 0.6 s after stage-1 burnout. The stack therefore carries the spent booster through the coast, which is the pessimistic reading.

Converged IV-1. Launch mass 592.2 kg, slant range 160.9 km at $t = 157.8$ s, intercept at 19.95 km and Mach 4.00, peak dynamic pressure 305 kPa. Dotted vertical rules mark pitchover, stage-1 burnout, separation and stage-2 burnout. The separation coast lasts 0.6 s, so it is a short grey segment. Source: runs/IV-1/converged.json.

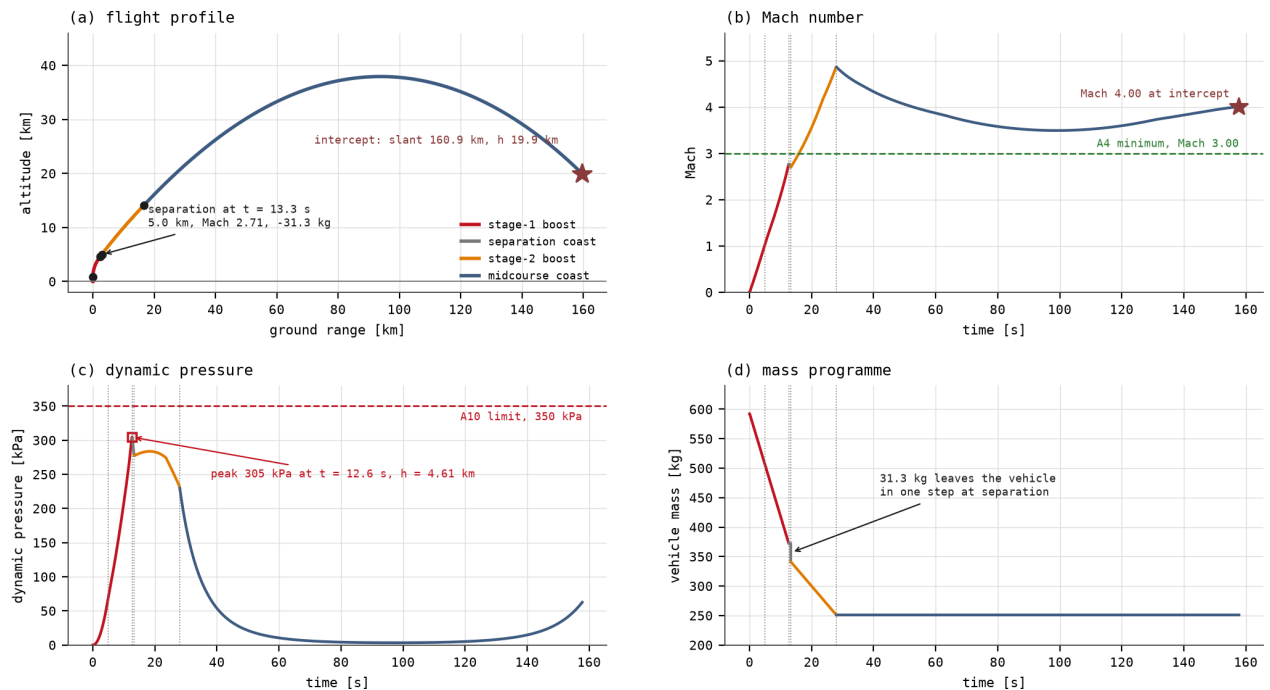


Figure 5. The converged ascent. Colour carries the phase. Panel (d) is the one a single-stage vehicle does not have: the mass drops by 31.3 kg in one step at separation.

Dotted lines are constant dynamic pressure, evaluated from the same US Standard 1976 table the trajectory used. The vehicle crosses its peak dynamic pressure early, under the booster, and arrives at the intercept with almost none. Source: runs/IV-1/figures/evidence_iv1.json.

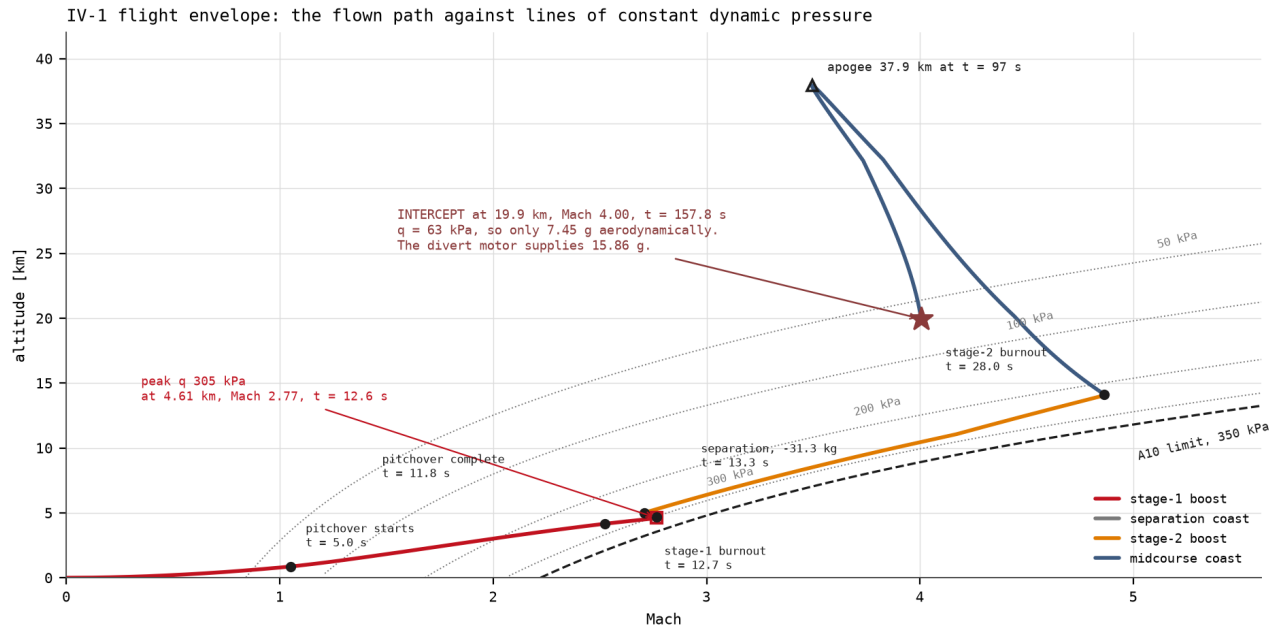


Figure 6. The same flight in altitude against Mach, against lines of constant dynamic pressure. The vehicle crosses its peak dynamic pressure early and under the booster. It then leaves the high-pressure region entirely, and arrives at the intercept where the air can supply only 7.45 g. That is why the divert motor exists.

The integrator reported: reached the 160.9 km slant range at $t = 157.80$ s, $h = 19.95$ km, Mach 4.00; available lateral acceleration 7.5 g is below the A11 minimum of 15.0 g

That warning is correct and is not a failure. It reports the AERODYNAMIC lateral acceleration only, because the integrator does not know about the divert motor. The A11 figure is the greater of the aerodynamic and the divert capability, and it is 15.86 g.

3. What makes IV-1 different from SV-1

SV-1 is a single body that keeps every kilogram it launches with. IV-1 is not. Three things change, and each of them changes the model rather than only the numbers.

3.1 Mass leaves the vehicle

At separation, 31.3 kg of spent booster and interstage leaves the vehicle in one step. That is the entire point of staging: the second propellant increment does not have to accelerate the first stage's empty structure.

The gain is measured against a controlled comparison, not asserted. Two vehicles fly the same 530 kg of propellant from the same 1,300 kg launch mass, at the same effective exhaust velocity, over the same 21.6 s. The burn time is identical, so the gravity loss of 211.8 m/s is identical too. The only difference is that the staged vehicle throws 120 kg away between the two increments.

Quantity	Staged	Equivalent single stage	Difference
Burnout speed [m/s]	1,117.3	1,046.1	+71.2
Burnout altitude [m]	16,633	10,129	+6504
Gain on burnout speed [%]			+6.80

Table 14. Staging is worth 6.80 percent on burnout speed at identical mass, propellant, burn time and therefore gravity loss. The closed form for the gain is v_e times the difference of the two logarithms of mass ratio. The integrated result matches it to $6.7e-13$ relative, so the jettison reaches the equations of motion exactly and not approximately.

3.2 The aerodynamic reference area changes at separation

A coefficient without its reference area is meaningless. The stack flies on the booster area of 0.1385 m^2 . After separation the payload stage flies on its own area of 0.0908 m^2 , which is 65.5 percent of it. Every drag and normal-force coefficient in this model therefore carries a stage index, and the aerodynamic model refuses to answer without one.

The reference length changes too, from 0.42 m to 0.34 m, so a static margin in calibres means a different distance before and after separation. That is why Table 12 evaluates the margin in both configurations rather than once.

3.3 The strakes carry their load by vortex lift

Each strake is 30 mm tall and 1.40 m long, so its exposed aspect ratio is 0.021. At that aspect ratio the flow is completely vortex dominated. A purely linear lifting-surface method would underpredict the load by two orders of magnitude.

The model therefore uses the Polhamus leading-edge suction analogy with the Lamar side-edge term. At the A11 angle-of-attack limit of 20 degrees, the strake normal force splits into 0.0348 of linear term and 0.3263 of vortex term. The vortex term is 90.4 percent of the load.

Mach	CN_max, strakes on	CN_max, strakes off	Gain [%]	x_cp shift [cal]
1.5	3.341	2.979	+12.1	0.055
2.0	3.341	2.979	+12.1	0.055
3.0	3.154	2.793	+12.9	0.082

Mach	CN_max, strakes on	CN_max, strakes off	Gain [%]	x_cp shift [cal]
4.0	3.022	2.661	+13.6	0.108
5.0	2.947	2.586	+14.0	0.125

Table 15. What the strakes are worth on the payload stage, at the converged dimensions. The gain on CN_max is +12.1 to +14.0 percent. On the starting design vector, which is the one the validation figure uses, the same gain is +13.5 to +15.8 percent.

On the stack the strakes move the centre of pressure forward by 0.135 to 0.138 calibres at the converged dimensions, and by 0.165 to 0.178 calibres at the starting dimensions. Forward is the wrong direction: strakes reduce static margin. On the payload stage alone the effect nearly cancels and changes sign with Mach, because the strake load centroid lands almost exactly on the centre of pressure that is already there.

IV-1 two-stage strake-stabilised aerodynamic build-up.
 Strake normal force by the Polhamus suction analogy with the Lamar side-edge term. References: NASA TN D-7921 Table III (Lamar and Gloss, 1975),
 a printed table; and NASA TM X-3130 (Jorgensen and Nelson, 1975), figures digitised. The inherited single-body physics is validated in fig_aero.py.

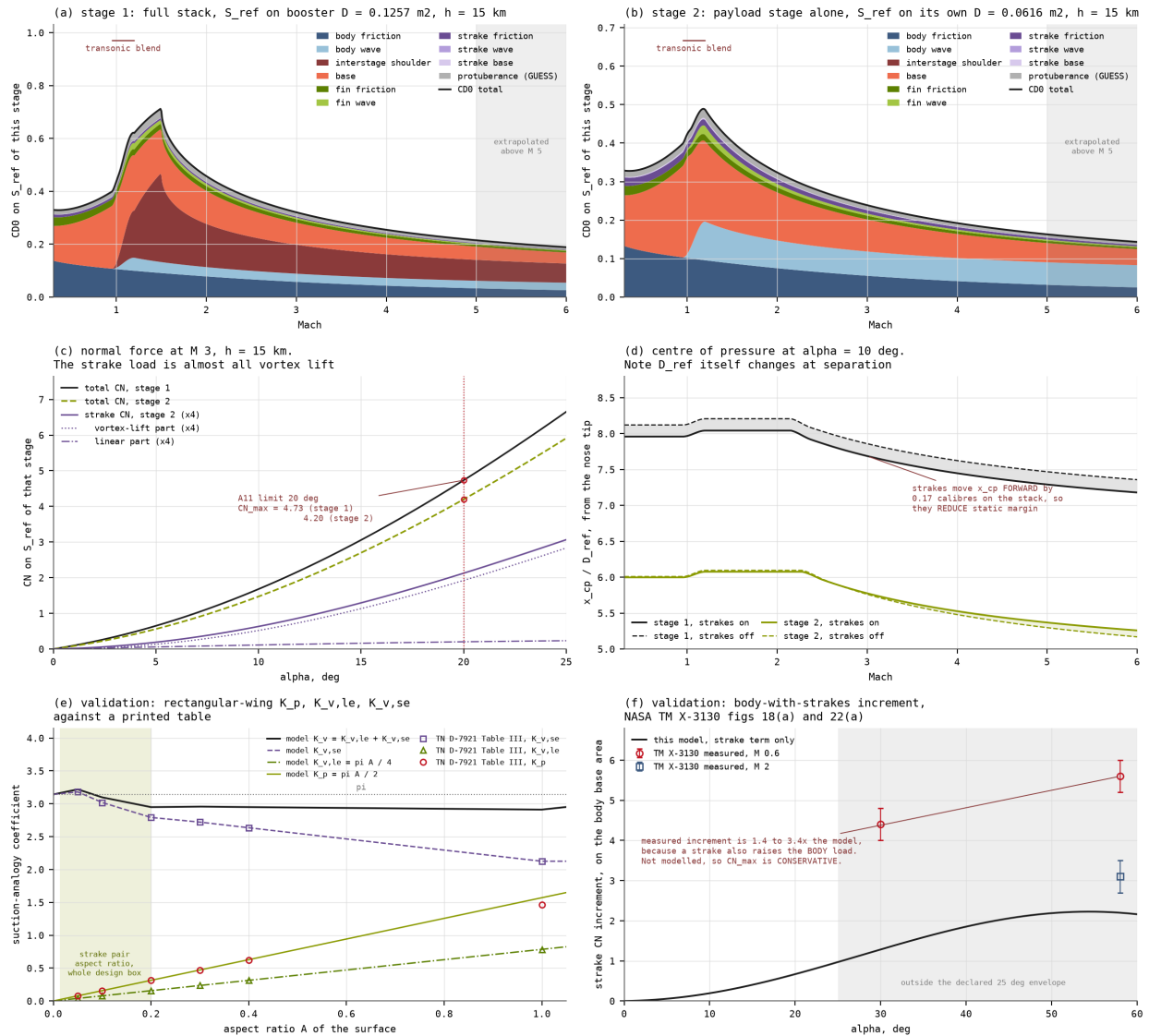


Figure 7. The strake and two-stage aerodynamic build-up, with the validation overlay. Panels (a) and (b) show that the same vehicle has two different drag polars, one per reference area. Panel (c) shows the strake load is almost all vortex lift. Panel (e) checks the suction-analogy coefficients against a printed table. Panel (f) checks the configuration increment against measurement.

The direction of the validation error matters, and it is favourable. Panel (e) reproduces the rectangular-wing coefficients of NASA TN D-7921 Table III, which is a printed table, so that check is exact. Panel (f) compares the model against the measured normal-force increment for a body with side strakes in NASA TM X-3130. **The measured increment is 1.4 to 3.4 times the model.** The reason is known: a strake also raises the load on the body itself, and that carryover is not modelled. The reported CN_{max} is therefore conservative, which is the safe direction for a sizing tool.

4. The nTop coupling

This section is the point of the exercise. nTop is not a downstream renderer here. It measures the solid it builds, and the measurements change the answer.

4.1 The notebook is authored programmatically

An nTop notebook is a binary container. It cannot be written by hand. The notebook is therefore emitted as recipe JSON and converted with an undocumented ntopcl subcommand:

```
ntopcl convert <recipe.json> <out.ntop> --dev-blocks-on=True
ntopcl exportjson <in.ntop> <out.json> --ext --dev-blocks-on=True
```

The IV-1 recipe is 422 blocks, built against a universe of 853 block signatures. It exposes 27 real notebook inputs, which is every geometry dimension of both stages, the interstage, the strakes and the fins. **One notebook therefore serves every design point:** the loop converts once and runs many times with different input JSON. That matters because conversion is expensive. Conversion evaluates the notebook, exports included.

Each stage is one closed profile revolved through 360 degrees. The strakes and the fins are extruded plates, patterned about the axis. Each airframe is then hollowed with an inward offset and its internal bays are subtracted, so each cavity is a real volume rather than an assumed fraction. The interstage is one cone block, which is a cylinder when the two stage diameters match, so the topology does not depend on a dimension.

The notebook measures four bodies separately: the booster, the payload stage, the interstage and the stacked union. The report and the mass model need them apart, because the booster and the interstage leave the vehicle and the payload stage does not.

4.2 Measurement accuracy

Every measured body is checked against an independent closed-form value derived from the design vector alone. The comparison is done at the starting design point, because that is the point whose closed-form reference was written to disk by the geometry study.

Body	Quantity	nTop	Closed form	Difference [%]
payload stage	volume	0.137458	0.137470	-0.008
payload stage	wetted area	2.079999	2.086290	-0.302
payload stage	strake wetted area	0.425257	0.427526	-0.531
booster	volume	0.263889	0.263894	-0.002
booster	wetted area	2.642237	2.638938	+0.125
interstage	volume	0.025686	0.025686	+0.000
interstage	wetted area	0.305411	0.305869	-0.150
stacked assembly	volume	0.431921	0.427049	+1.141
stacked assembly	wetted area	5.027648	5.031097	-0.069
stacked assembly	strake wetted area	0.425257	0.427526	-0.531

Table 16. nTop measurements against independent closed-form geometry. The three separate bodies agree to better than 0.01 percent on volume and better than 0.31 percent on wetted area. The stacked-assembly volume differs by 1.14 percent for a known reason: nTop measures the union, and the

union includes the strake and fin solids that the closed form does not add to the body volume.

The strake wetted area was confirmed three independent ways. First, the measured area of 0.425257 m^2 agrees with the closed form for a solid plate to -0.531 percent. Second, the measured strake solid volume of 0.00134475 m^3 agrees with the plate volume to +0.056 percent. Third, halving the strake height scales the measured area to 60.3 percent and the volume to 50.1 percent, which is what a plate of half the height must give once its unchanged edge faces are counted.

The measured solid area is 1.2724 times the zero-thickness reference of two times n times height times length. That is not an error. An 8 mm plate 30 mm tall is 27 percent edge, and skin friction acts on the real surface, so the measured number is the right one to hand to the aerodynamic model.

4.3 The coupling changes the answer

The same design vector was sized twice: once with analytic geometry, and once with the geometry nTop measured.

Quantity	Analytic geometry	nTop-measured	Change
Launch mass [kg]	585.6	592.2	+6.6
Intercept altitude [km]	17.6	19.9	+2.3

Table 17. The same design vector with and without measured geometry. The measured two-stage airframe is heavier than the analytic shell estimate, so the launch mass rises. The analytic figures are quoted from `examples/IV-1/README.md`, which records the analytic run; only the measured run is stored as JSON.

12.5 percent of the launch mass is nTop-measured: 39.96 kg for the booster airframe and 24.17 kg for the payload stage with its strakes and fins. The fraction is small for a structural reason, not a modelling one. The payload is a requirement, the propellant is a design variable, and the motor inert masses come from a correlation. None of those three is geometry, and together they are most of the vehicle.

4.4 What a measurement costs

Run	Wall time [s]
baseline	54.2
small strakes	55.2
alternate	55.4
area distribution	69.0
converged	58.1

Table 18. Measured wall time of the five stored measurement runs. A separate controlled repeat of five identical jobs measured 55.0, 78.6, 92.7, 114.7 and 117.8 s. The two-times spread on repeats of an identical job is real and is not attributed here. Budget the upper end.

Per-block timings say where the time goes, so the choice of what to measure is informed rather than guessed. Four surface-area calls on implicit bodies dominate: 24.6 s on the stage-2 body, 24.2 s on the stage-2 fins, 23.8 s on the strakes and 16.9 s on the stage-1 fins. The fifth, on the booster's plain cylinder, costs 0.27 s. Mass properties cost 15.0 s on the stage-2 structure and 8.6 s on the stack union. **Nothing in that list is meshing.** The

mesh is used only for the STL export and never for a measurement.

A build ladder confirms the accounting. Building only the stage-2 outer mould line costs 14.7 s, adding its plates 42.2 s, hollowing it 50.5 s, adding the booster 76.3 s and the full stack 117.8 s. Each added body costs what its own area and mass blocks cost, and nothing more.

5. Verification

552 automated tests pass and 1 is skipped. The suite takes 9.6 minutes, because parts of it drive real ntopcl subprocesses. The command and its recorded output are in runs/IV-1/pytest.txt.

```
.venv/Scripts/python.exe -m pytest tests -q
```

Vehicle	Test modules	Tests collected
SV-1, the regression baseline	7	296
IV-1, new in this work	5	257
Total	12	553

Table 19. Test counts by vehicle. SV-1 is the regression baseline and its 296 tests are unchanged by this work.

5.1 The new physics reduces to the validated physics

The strongest check on a generalisation is that it reproduces the thing it generalises. A one-stage stack is compared against the validated SV-1 single motor. Sixteen operating-point quantities are compared with exact equality, not with a tolerance: propellant mass, chamber and exit pressure, throat and exit area and diameter, burning area, the Kn ratio, burn rate, mass flow, burn time, vacuum and sea-level thrust, the thrust coefficient and the vacuum specific impulse.

The thrust curves are also compared. The difference is exactly 0.0 N over 20001 samples at three altitudes, while both models are inside the boost shape. After that point the SV-1 motor transitions to a phase a single stage does not have, so no tolerance would be meaningful there and none is claimed.

5.2 Closed-form cases

Case	Relative error
Staged Tsiolkovsky with jettison, against the closed form	2.4e-08
Gain of staging against the closed-form gain	6.7e-13
Mass bookkeeping, m0 less burned less jettisoned	7.5e-15
Specific-energy drift with no thrust and no drag	5.3e-15
Vertical vacuum climb apogee, against $h_0 + V_0^2/(2g)$	9.0e-14
Fast atmosphere lookup against numpy.interp	1.4e-15

Table 20. Cases with an analytic answer. The staged, mass-bookkeeping and energy figures are the stated assertions of tests/test_trajectory_iv1.py. The staging-gain and atmosphere figures are measured by rocketgen/report/evidence_iv1.py. Fourth-order Runge-Kutta convergence was checked separately: halving the step reduces the error by 15.93 and 16.01, against the factor of 16 the method requires.

The atmosphere extension is validated against the published layer breaks, with the geopotential conversion done properly. The fast uniform-grid lookup that replaced `numpy.interp` is bit-identical to it, at $1.4\text{e-}15$ relative over the whole range, and costs 2.30 microseconds per call. The table reproduces the stratopause maximum of 270.65 K, which matters because a lofted midcourse spends its time there.

5.3 One test is skipped, not passing

The suite is not clean, and this report will not present it as if it were. One test is skipped. It covers the reporting branch that fires when a trajectory flies above the atmosphere-table ceiling. That branch is now unreachable on any real trajectory, because the table reaches 86 km and the highest measured arc apogees at 54 km. The mission module reads the ceiling directly from the atmosphere module, so there is no seam through which a test can drive the ceiling low. The test therefore skips with that reason recorded, rather than being deleted.

6. Fidelity and limitations

This is a Class-I conceptual tool. It is useful for choosing between configurations. It is not a substitute for computational fluid dynamics, for structural analysis or for six-degree-of-freedom flight mechanics.

The vehicle and its requirements are invented for this demonstration. They correspond to no real programme. Nothing here describes guidance, seekers, warheads, energetics or countermeasures. Section 1.1 shows the requirements were not even self-consistent when written.

6.1 The largest unquantified optimism

Nozzle flow is ideal. The model assumes frozen composition, constant specific-heat ratio and single-phase flow. It has no two-phase alumina loss, no divergence loss, no combustion inefficiency and no throat erosion. Real delivered specific impulse for this propellant class is typically 3 to 7 percent lower. **That penalty is not applied, because its magnitude could not be sourced.** Slant range and intercept Mach are therefore optimistic by an amount that has not been quantified. This is the single largest known optimism in the result.

6.2 Limitations specific to this vehicle

- **The 350 kPa dynamic-pressure limit is not verified by the structural model.** Structural sizing is wall thickness times density, plus a hoop-stress check on each motor case and a buckling check on the interstage. Nothing sizes the airframe for a 350 kPa aerodynamic load, so the airframe mass is optimistic by an amount this toolkit cannot quantify.
- **Separation is instantaneous.** It imparts no impulse and no attitude disturbance, and it has no drag transient. The separation joint is not costed: a real tandem separation needs a linear shaped charge or a clamp band, springs or retro-rockets, and ring frames at both ends of the interstage. The jettisoned mass is therefore optimistic.
- **There is no thrust-vector control.** The divert motor is a static capability figure at the intercept condition. It says nothing about response time, actuator rate, plume interaction, minimum impulse bit, roll coupling or autopilot behaviour.
- **Strake-to-fin vortex interference is not modelled.** The interference factor is 1.0, which means not modelled rather than measured as unity. No factor for this configuration was sourced.
- The trajectory is a three-degree-of-freedom point mass on a flat earth with constant gravity. There is no roll, no sideslip and no autopilot.

- The grain geometry is a neutral-burning tube closed at its mean web. There is no burnback simulation, so the reported grain dimensions are equivalent values rather than a drawing.
- Requirement A9, the static margin, is not among the constraints the sizing script records. Section 2.2 evaluates it and it does not pass. That is an open item.

6.3 The pitch programme, measured

The pitch programme is open loop and has no autopilot gain of any kind. The vehicle rises vertically until $t = 5.0$ s, turns at the commanded rate of 8.0 deg/s until the flown flight-path angle reaches the commanded angle, then flies a pure gravity turn. The commanded rate is a design variable, not a gain on an angle error.

With no thrust-vector control, that commanded rate is often beyond the angle-of-attack limit, and the module says so. **On this design point it is not.** The turn finished at $t = 11.76$ s, which is an average flown rate of 7.70 deg/s against the 8.0 deg/s commanded. The peak angle of attack was 8.71 degrees against the 20 degree limit, and 0 steps hit the limit. The turn starts at 5 s, where the dynamic pressure is already high, which is why the airframe can fly it. An earlier or higher pitchover would be limited, and the diagnostic would record it.

6.4 The drag calibration is not applied to this vehicle

The SV-1 aerodynamic build-up runs about 15 percent low on zero-lift drag against 23 Basic Finner free-flight shots, so the SV-1 sizing loop scales drag by 1.171 at its boundary. **The IV-1 sizing script does not apply that correction.** It builds the aerodynamic model directly and does not wrap it in the calibrated boundary. The IV-1 drag is therefore the uncorrected physics, which is low, and low drag overpredicts range. The direction of this error is unfavourable and it is not quantified for the two-stage configuration.

6.5 Every value that is a guess

Every empirical constant in the code is registered with its source. A value that is a guess must say so in its source text, and a test asserts that. The table lists every registered source whose text admits a guess, a modelling choice, an approximation or an assumption.

55 of 145 registered sources are flagged. The registry is global, and modules register into it when they are imported. The count is therefore only complete if every module is imported before the registry is read. All five IV-1 modules are imported first, then the shared physics modules they build on, then the notebook-authoring modules. The module column below says which module owns each entry.

Module	Count
propulsion	14
propulsion_iv1	9
config_iv1	5
trajectory	5
trajectory_iv1	5
aero_iv1	4
masses	4
aero	3
masses_iv1	3

Module	Count
ntopgen.stack_notebook	2
ntopgen.rocket_notebook	1
Total flagged	55

Table 21. Flagged sources by owning module. The shared physics modules are counted because IV-1 reuses them unchanged. A few of their entries describe the SV-1 dual-thrust motor and the SV-1 dive, which IV-1 does not fly; they are listed rather than filtered, because filtering a source registry by hand is how a limitation goes missing.

Registered name	Module	Why it is not a measured value
aero_isentropic_compression_turn	aero	Pressure rise through a compressive turn taken as an exact isentropic Prandtl-Meyer compression, $nu_2 = nu_1 - \Delta$, with $C_p = (2/(\gamma M_1^2))(p_2/p_1 - 1)$. Same gas dynamics and the same reference as the boattail expansion, Liepmann and Roshko, Elements of Gasdynamics (1957), run in the opposite direction. APPROXIMATION: the entropy rise across the real oblique shock is not modelled, and when Δ exceeds the available Prandtl-Meyer angle the shock detaches and the value is clamped at the sonic result, which UNDERSTATES the detached-shock pressure. Used by `aero_iv1` for the interstage shoulder; the single-body SV-1 build-up has no compressive turn and so does not exercise it.
aero_nozzle_exit_fraction_GUESS	aero	GUESS. Used only when `area_nozzle_exit` is None and `power_on=True`: the nozzle exit is assumed to occupy $A_e/A_{base} = 0.50$. This is a placeholder for the real value, which WP3's motor model must supply. It changes only the powered-base drag relief.
aero_protuberance_GUESS	aero	GUESS. A single lumped allowance for antennas, umbilicals, fin-root fairings, lugs, joints and surface imperfections, taken as 5 percent of the sum of the other zero-lift components. There is no correlation behind this number. It appears in the breakdown as 'CD_protuberance_GUESS' so it can never be mistaken for a computed quantity.
bulkhead_allowance	masses	GUESS: four ring bulkheads, each a full-diameter disc of 40 percent of wall thickness. Not from a source; a placeholder for joint, rail and lug hardware
case_minimum_gauge	masses	Manufacturing minimum gauge, not a strength requirement. Hoop stress alone gives a sub-millimetre wall at this chamber pressure, which cannot be filament wound or machined. Floors of 1.8 mm (composite, roughly 3 to 4 wound plies) and 1.2 mm (metallic) are taken as engineering minimums. GUESS: no public minimum-gauge standard was found, but without the floor the model returns a motor mass fraction of 0.95, well above the 0.80 to 0.92 band for real tactical motors
insulation_thickness	masses	Sutton and Biblarz, Rocket Propulsion Elements, 9th ed., Chapter 14: EPDM chamber insulation 1 to 5 mm for short-burn tactical motors; 2.5 mm taken here as a mid-range modelling choice

Registered name	Module	Why it is not a measured value
iv1_acs	config_iv1	Attitude-control motor performance. Isp 235 s is the low end of the AP/HTPB range in Sutton and Biblarz, Rocket Propulsion Elements 9th ed., Table 12-1, taken low because a side thruster runs a short, low-expansion nozzle at altitude and pays a large divergence and heat loss. GUESS: the 0.55 inert-to-propellant ratio for a pulsed divert pack is not from a source; it is high relative to a single large motor because a thruster bank pays for multiple valves, nozzles and a manifold. No plume interaction, no minimum impulse bit, no control loop and no roll coupling are modelled.
iv1_aero_fin_station	aero_iv1	MODELLING CHOICE, not a measurement: each stage's tail fins are placed with their root TRAILING EDGE at the aft end of that stage, so the stage-1 fins end at the booster base and the stage-2 fins end at the stage-2 to interstage joint. `StageSpec` carries no fin station, and SPEC_IV1.md section 3 shows tail fins at the aft end of each stage. Exposed as the constructor argument `fin_te_gap` so a real gap can be set. The consequence for stability is that the stage-2 fins are about one interstage length further forward in the stacked configuration than the booster fins, which is exactly the geometry the static-margin check needs.
iv1_aero_strake_Kv	aero_iv1	K_v, the vortex-lift factor, is the SUM of a leading-edge and a SIDE-EDGE term, $K_v = K_{v,le} + K_{v,se}$, because a strake is rectangular and unswept: its shed vortex comes off the streamwise tip edge, not off a swept leading edge. This is the Lamar-Polhamus extension of the suction analogy to non-delta planforms: J. E. Lamar, 'Prediction of Vortex Flow Characteristics of Wings at Subsonic and Supersonic Speeds', NASA TR R-428 (1974); J. E. Lamar and B. B. Gloss, NASA TN D-7921 (1975). $K_{v,le} = \pi A/4 = K_p/2$, which TN D-7921 Table III reproduces to better than 0.1 percent for every rectangular entry from $A = 0.05$ to $A = 0.40$. $K_{v,se}$ is taken directly from the TN D-7921 Table III rectangular-wing column (the m1 column, the TR R-428 continuous-loading method that the authors take as the standard), interpolated in A and anchored at $A = 0$ by the analytic limit $K_{v,se} \rightarrow \pi$ stated in TN D-7921 p. 9 and TR R-428 p. 19. See LAMAR_KV_SE_RECTANGULAR for the table. At the default strake pair aspect ratio of 0.043 that gives $K_v = 3.19$. NOTE, because it matters: the value is close to π, NOT to the 2.0 that Bollay's zero-aspect-ratio nonlinear wing theory (ZAMM 19, 1939) gives and that the Newtonian '$2 \sin^2 \alpha$' term in SOURCES['aero_fin_normal_force'] uses. The suction analogy is about 60 percent higher. That is exactly why a strake needs its own method instead of being pushed through the inherited fin routine. APPROXIMATIONS, both of which push the strake normal force DOWN: the interpolation is linear in A, so it is C0 but not C1 with respect to the strake design variables (it is exactly constant with respect to Mach and alpha, so the trajectory integrator is unaffected); and a body-mounted strake sits in the accelerated cross-flow around the body, where the real vortex is stronger than on the isolated plate the table describes. OMITS vortex breakdown, all Mach dependence, and the strake-to-strake interaction of a cruciform set.

Registered name	Module	Why it is not a measured value
iv1_aero_strake_drag	aero_iv1	Strake drag reuses the fin machinery where the physics is identical and says where it is not. Friction: the same Sommer and Short compressible flat-plate law and the same Raymer thickness form factor as the fins, with the Reynolds number on the strake LENGTH, because the strake length is its streamwise chord. Identical physics, so SOURCES['aero_skin_friction'] and SOURCES['aero_fin_form_factor'] apply unchanged. Thickness wave drag: the same Ackeret double-wedge result, SOURCES['aero_fin_wave_drag']. NOT identical physics, and the difference is declared: a real strake is a constant-thickness plate with a short leading-edge bevel, not a double wedge peaking at mid-chord. Concentrating the same thickness into a bevel a few thicknesses long raises the local surface slope by more than an order of magnitude and the wave drag with it, so the double-wedge assumption UNDERSTATES strake wave drag. The opposite end is reported as `xcheck_CD_strake_le_blunt_bound`: stagnation pressure on the whole strake frontal area $n \cdot \text{thickness} \cdot \text{height}$, which is what a fully blunt unbeveled leading edge would cost. The two BRACKET the truth, and at the default geometry the bracket is wide, more than two orders of magnitude, because the double-wedge t/c of 0.0057 spread over a 1.4 m chord is an almost flat surface while a blunt edge is not. Where the real strake falls inside that bracket depends on the leading-edge bevel length, which is NOT an IV-1 design variable, so this model reports the optimistic end plus the bound rather than interpolating with an invented bevel length. Trailing-edge drag: the strake trailing edge is blunt and its base area is charged with the same base-pressure correlation as the body base, SOURCES['aero_base_drag']; that correlation was fitted to an AXISYMMETRIC body base, so applying it to a small two-dimensional trailing edge is an EXTENSION, taken because leaving the term out entirely is the larger error.
iv1_aero_strake_upwash	aero_iv1	The Barrowman body-upwash factor $1 + a/s$ (a = body radius, s = strake tip radius) is applied to the strake POTENTIAL term only, reusing SOURCES['aero_fin_body_upwash']. It is a linear potential-flow result, so it is not applied to the vortex-lift term: there is no published upwash factor for a separated-flow load, and the suction analogy already integrates the separated load. APPROXIMATION with a known sign: the real cross-flow acceleration around the body raises the effective incidence of the strake panel, so leaving the vortex term un-augmented makes the strake normal force a LOWER BOUND. Because the potential term is negligible at strake aspect ratios, this choice moves the answer by well under 1 percent.
iv1_fin_te_at_stage_aft_end	ntopgen.stack_notebook	MODELLING CHOICE, and it is visible in the render. SPEC_IV1 gives `StageSpec` a fin semi-span, root chord, taper, sweep and thickness but NO longitudinal station, so the trailing edge is placed flush with each stage's own aft end, which is what 'tail fin' means. On stage 2 that puts the panels immediately forward of the interstage, and because the stage-2 tip-to-tip span (0.56 m on the default) exceeds the booster diameter (0.40 m) the panels stick out past the interstage flare. A real vehicle would move them forward or use a wrap-around skirt. Nothing in the aero or mass model depends on the station, so this is a picture-level simplification, not a numerical one.

Registered name	Module	Why it is not a measured value
iv1_interstage_buckling	masses_iv1	The interstage is a compression member: it transmits the full booster thrust into the upper stage. Classical buckling of a thin unstiffened cylindrical shell under axial compression is $\sigma_{cr} = E \cdot t / (R \cdot \sqrt{3 \cdot (1 - \nu^2)})$, which is $0.605 \cdot E \cdot t / R$ at $\nu = 0.33$. Real shells reach only a fraction of that because of imperfection sensitivity. A knockdown factor of 0.30 is applied, which is the conservative end of the 0.15 to 0.50 band quoted for unstiffened monocoque shells in NASA SP-8007, Buckling of Thin-Walled Circular Cylinders. GUESS: 0.30 was chosen inside that band rather than computed from an imperfection survey, and the shell here is conical rather than cylindrical, so the check is an approximation applied at the smaller radius, which is the critical end.
iv1_interstage_mass	masses_iv1	MODELLING CHOICE: the interstage is a truncated conical shell between the two stage diameters, of thickness $dv.t_{interstage}$ in aluminium 7075-T6, plus two ring frames at 40 percent of the shell thickness. Separation hardware (explosive bolts or a linear separation joint, springs or pistons, and the electrical disconnect) is NOT costed. GUESS: that omission makes the jettisoned mass optimistic by an unquantified amount.
iv1_lateral_accel	config_iv1	Available lateral acceleration is the static value $q \cdot S_{ref} \cdot CN_{max} / m$ at the intercept condition, with CN_{max} taken at the alpha limit. APPROXIMATION: it is a capability, not a manoeuvre. It says nothing about autopilot response time, actuator rate or the transient during a turn.
iv1_ogive_polygon_sampling	ntopgen.stack_notebook	Inherited, not re-guessed. The revolved outer mould line is a chord polygon, so its volume is exactly the frustum sum that `rocketgen.sizing.masses._tangent_ogive_volume` computes with the same n. WP4 swept n on the SV-1 nose against a 20000-segment reference and measured a total outer-mould-line volume error of -0.106 percent at n=8, -0.0117 percent at n=24 and -0.0004 percent at n=128. The IV-1 stage-2 nose is SHORTER and BLUNTER than SV-1's (1.008 m on a 0.28 m body, fineness 3.6, against 1.05 m on 0.35 m at fineness 3.0), and a chord polygon's error scales with the curvature it has to cut across, so n = 24 is at least as accurate here. It is kept unchanged so the two notebooks are comparable.
iv1_qmax	config_iv1	MODELLING CHOICE: 250 kPa dynamic-pressure limit, against 200 kPa for SV-1. A vertical sea-level launch accelerates through dense air, so the same airframe sees a higher peak than SV-1 does launching at 10 km. Consistent with the manoeuvre-envelope pressures quoted for supersonic tactical vehicles in Fleeman, Tactical Missile Design, 2nd ed., Chapter 3.
iv1_qmax_revision	config_iv1	MODELLING CHOICE, and a declared gap. A10 was revised from 250 kPa to 350 kPa because a sweep over propellant, thrust, pitchover angle and pitchover time found a hard floor of about 278 kPa on peak dynamic pressure, rising to 309 kPa for any design that also meets A2, A3 and A4. The structural model does not size the airframe for that load: it is wall thickness times density plus a motor-case hoop-stress check and an interstage buckling check. The airframe mass is therefore optimistic by an amount not quantified here.

Registered name	Module	Why it is not a measured value
iv1_separation	config_iv1	MODELLING CHOICE: stage separation is instantaneous, imparts no impulse and no attitude disturbance, and jettisons the stage-1 inert mass together with the interstage. Real separation carries a tip-off disturbance and a brief drag transient; neither is modelled.
iv1_strake_mass	masses_iv1	Strake mass is the swept plate volume times the titanium density in MATERIALS['fin_ti64'], with a 0.85 area-to-volume factor for a tapered leading edge and a root fillet. GUESS: the 0.85 factor is not from a source; a constant-section plate would be 1.0 and a full double wedge 0.5, so it sits between them.
ogive_polygon_sampling	ntopgen.rockett_notebook	Measured, not guessed. The revolved OML is a chord polygon, so its volume is the frustum sum that <code>`rocketgen.sizing.masses._tangent_ogive_volume`</code> computes with the same n. Sweeping n against a 20000-segment reference on the default SV-1 gives a TOTAL OML volume error of -0.106 percent at n=8, -0.026 percent at n=16, -0.0117 percent at n=24, -0.0066 percent at n=32 and -0.0004 percent at n=128. n=24 is used for the outer profile: 85x inside the 1 percent gate in PLAN.md, and small enough to keep the recipe near 300 blocks.
prop.burn_rate_coefficient	propulsion	MODELLING CHOICE inside a cited range: reference burn rate 10.0 mm/s at 1000 psia. Sutton Table 12-1 (HTPB/AP/Al) gives 0.25 to 3.0 in/s = 6.35 to 76.2 mm/s at 1000 psia, and the Ch.12 text on p.476 states most composite propellants burn at 7 to 20 mm/s. The coefficient a follows from $a = r_{ref} / p_{ref}^n$.
prop.case_safety_factor	propulsion	MODELLING CHOICE: safety factor 1.5 applied to material yield strength in the thin-wall hoop-stress case sizing ($\sigma = p \cdot r / t$), plus a 1.0 mm minimum practical gauge. 1.5 is the conventional aerospace pressure-vessel factor; it was not taken from a specific standard in this session.
prop.gamma	propulsion	ASSUMPTION (not tabulated, treat as a guess inside a physically bounded band): ratio of specific heats of the combustion products taken as 1.20, the value normally used for aluminized AP/HTPB exhaust. Cross-checked: with gamma = 1.20 the cited Isp (262.5 s) and flame temperature (3440 K) imply a mean exhaust molar mass of about 26 kg/kmol, inside the 25 to 30 kg/kmol band of Sutton Figure 12-3 for HTPB-based composite propellant at 68 atm. A +/- 0.02 change in gamma moves the derived c* by about 1.5 %.
prop.ideal_nozzle	propulsion	APPROXIMATION: frozen-composition, constant-gamma, single-phase isentropic nozzle flow. No two-phase Al2O3 loss, no divergence loss, no combustion efficiency and no throat erosion. Reported Isp is therefore an ideal value; real delivered Isp for a tactical motor of this class is typically 3 to 7 % lower (magnitude of that correction NOT sourced here, so it is not applied).
prop.igniter_mass	propulsion	GUESS: igniter and safe-arm mass taken as 0.5 % of total propellant mass.
prop.insulation_thickness	propulsion	GUESS: 3.0 mm of EPDM internal insulation on the case wall and domes. Real insulation is sized from local gas-side heat flux and exposure time, which is out of scope (SPEC.md section 8). Density from MATERIALS['insulation_epdm'].

Registered name	Module	Why it is not a measured value
prop.motor_mass_fraction	propulsion	CROSS-CHECK CORRELATION, quoted band only, treat the endpoints as a guess: tactical solid rocket motors have propellant mass fractions $m_p / (m_p + m_{\text{inert}})$ of roughly 0.80 to 0.91. This band was not verified against a primary source in this session; it is used only to bound the physics-route inert mass, never to replace it silently.
prop.neutral_burning	propulsion	APPROXIMATION: neutral (constant) burning is assumed within each phase. The burn-rate law is closed at the MEAN web position, which makes propellant mass, web thickness, mass flow and burn time mutually consistent ($w = V_p / A_{b_mean}$, $t_b = w / r$, $\dot{m} = \rho * A_{b_mean} * r$). No burnback simulation is performed, so the reported grain dimensions are volume- and area-equivalent, not a manufacturing drawing.
prop.nozzle_mass_model	propulsion	GUESS-LEVEL geometric estimate: conical nozzle, 15 deg half angle, 8.0 mm wall, bulk graphite / carbon-carbon density 1800 kg/m ³ (handbook range 1.7 to 1.9 g/cm ³ , not verified against a datasheet in this session), multiplied by 1.5 to allow for the convergent section, throat insert and attach ring. This estimate is known to be optimistic; compare against SOURCES['prop.motor_mass_fraction'].
prop.ramp_times	propulsion	MODELLING CHOICE, not physics: ignition rise 0.05 s (specified by the WP3 task statement), boost-to-sustain blend 0.10 s, tail-off 0.20 s. The ramps are mass-conserving by construction so total impulse is unchanged.
prop.separation_criterion	propulsion	Summerfield flow-separation criterion, separation when p_e / p_a is below about 0.4 (Summerfield, Foster and Swan, 1954; the 0.4 figure is the commonly quoted approximation and was NOT verified against the primary paper in this session). Used only to raise a warning, never to change a force.
prop.terminal_grain	propulsion	MODELLING CHOICE: the terminal segment is an internal-burning tubular grain, the same geometry and the same mean-web closure as the boost segment, because an end burner at this bore cannot supply the burning area a multi-kilonewton terminal thrust needs at a credible chamber pressure. Its length counts towards the SPEC.md section 4 grain L/D limit of 1.0 to 8.0.
prop.terminal_pulse	propulsion	MODELLING CHOICE with one guess: the terminal pulse is a third grain segment in the same case, ignited on command rather than on a timer, which is how real dual-pulse tactical motors work (an inter-pulse bulkhead separates the charges and a second igniter fires the aft one). Its default chamber pressure is $1.0 \times \text{DesignVector.p}_c$, that is the pressure the case is already sized for, so no new number is introduced. GUESS: the inter-pulse bulkhead, its insulation and the second igniter are NOT costed as separate inert mass; the terminal grain is charged only the same 0.5 % igniter fraction as the rest of the motor, so the reported inert mass is optimistic for a pulsed motor. Physical motivation: an unpowered terminal dive is terminal-velocity limited and cannot reach Mach 1.5 at sea level for any dive angle, so thrust in the endgame is the only way to meet SPEC R6.

Registered name	Module	Why it is not a measured value
prop.two_position_throat	propulsion	MODELLING CHOICE: the motor is given a two-position throat (ejectable boost throat insert) so the boost and sustain phases can run at different throat areas with the same nozzle area ratio. Without it, a single throat sized for the boost thrust drives the sustain chamber pressure to about 0.13 MPa and Kn to about 22, which no composite propellant will sustain. Set two_position_throat=False to model one fixed throat instead.
prop_iv1.booster_aft_closure	propulsion_iv1	MODELLING CHOICE: the booster grain bay is the stage length less a 0.10 m aft closure, nozzle attach ring and igniter boss allowance. The value is not new: it is the same CASE_LENGTH_ALLOWANCE_M that propulsion.py already adds to the grain length to get the case length, so bay length and case length are consistent with each other. The payload stage instead gets $L - L_{nose} - L_{seeker} - L_{payload_bay}$, which is set by the internal layout in config_iv1.StackDesignVector.
prop_iv1.fin_mass_model	propulsion_iv1	GUESS, carried over unchanged from the analytic branch of masses.py::build_masses so the two modules cannot disagree about the same hardware: fin mass is $n_{fin} * S_{fin_exposed} * t_{fin} * 0.65 * \text{density}(\text{Ti-6Al-4V})$, where 0.65 is an area-to-volume factor for a tapered biconvex section. No source was found for the 0.65; it is a shape factor chosen to be below the 1.0 of a flat plate.
prop_iv1.interstage_shell	propulsion_iv1	MODELLING CHOICE: the interstage is charged as a cylindrical shell at the BOOSTER diameter, of length StackDesignVector.L_interstage and thickness t_interstage, in MATERIALS['airframe_al7075']. Mass is then exactly $\pi * D1 * L * t * \rho$, linear in L_interstage. The conical frustum from D2 to D1 is computed alongside it as a cross-check and is lighter, because a taper has less lateral area than a cylinder of the same axial length; the heavier cylindrical value is the one charged. GUESS: no buckling or crush-load sizing is performed, so t_interstage is an input, not a result.
prop_iv1.p_c_min_reliable	propulsion_iv1	GUESS: 1.0 MPa taken as the chamber pressure below which composite-propellant combustion becomes unreliable, so a stage below it is warned about. The figure is the same one propulsion.SolidMotor already warns on and no primary source for it was verified in this session. It changes no force and no mass: it only raises a warning. Every IV-1 stage runs at StageSpec.p_c, which defaults to 8.0 MPa, so the check is inactive for the default stack and exists for design vectors the sizer may reach.
prop_iv1.separation_hardware	propulsion_iv1	GUESS, and an admitted omission: the separation joint itself is NOT costed. A real tandem separation needs a linear-shaped-charge or clamp-band joint, springs or retro-rockets, and ring frames at both ends of the interstage. None of that appears in jettisoned_mass(), so the reported jettisoned mass is the shell and motor hardware only and is optimistic by an amount this model cannot quantify.

Registered name	Module	Why it is not a measured value
prop_iv1.separation_timing	propulsion_iv1	MODELLING CHOICE, following the deliverable contract: stage 1 ignites at $t = 0$ and burns to the end of its tail-off, the stack then coasts unpowered for <code>InterceptRequirements.t_coast_separation</code> , and separation and stage-2 ignition are the same instant, $t_{\text{separation}} = t_{\text{burnout}}(1) + t_{\text{coast_separation}}$. <code>SPEC_IV1.md</code> section 5 allows a short unpowered coast between boost and stage-2 ignition and does not fix whether the jettison happens at the start or the end of it; putting it at the end means the stack carries the booster mass through the coast, which is the pessimistic and therefore the safe choice for the trajectory.
prop_iv1.stage1_sized_at_sea_level	propulsion_iv1	MODELLING CHOICE, following the <code>StageSpec.F_thrust</code> definition in <code>config_iv1.py</code> : stage-1 thrust is a SEA-LEVEL equivalent, because <code>SPEC_IV1.md</code> requirement A1 launches the stack vertically from a canister at sea level. The stage-1 throat is therefore sized with $p_a = 101325 \text{ Pa}$. Upper-stage thrust is a VACUUM value and the upper-stage throats are sized with $p_a = 0$, because they are lit above 10 km. Both use the same relation $F = C_F \text{vac} * p_c * A_t - p_a * A_e$; only the sizing p_a differs.
prop_iv1.stage_is_a_phase	propulsion_iv1	MODELLING CHOICE: one stage of the IV-1 stack is modelled with exactly the same closure as one phase of the SV-1 dual-thrust motor, that is one propellant charge at one chamber pressure through one fixed throat, sized by <code>propulsion.design_phase()</code> . The physical difference between a phase and a stage is not in the internal ballistics but in the hardware: a stage brings its own case and nozzle, and stage 1 leaves the vehicle at separation.
prop_iv1.tubular_stage_grain	propulsion_iv1	MODELLING CHOICE and a known conservatism: every stage grain is an internal-burning tube closed at its mean web, the same geometry as the SV-1 boost segment. A tube is the least area-efficient internal-burning geometry. A slotted, star or finocyl grain reaches several times the burning area in the same case length, which is how real boosters of this thrust class stay short. No shape factor is applied here, because no sourced multiplier was available; the model reports the long tube and its L/D instead. A stage that fails the <code>SPEC_IV1.md</code> section 6 L/D limit of 1.0 to 8.0 with a tubular grain may still close with a shaped grain, and the warning says so.
radome_thickness	masses	GUESS: 6 mm Pyroceram radome wall. No public thickness standard was found for a 0.35 m class radome; scaled from the requirement that wall thickness be an integer number of half-wavelengths at Ku band, which is not modelled here
traj.alpha_limit	trajectory	MODELLING CHOICE: trim angle of attack is limited to 15 deg. Trim demands beyond that are recorded as an authority shortfall rather than flown, so the sizing loop can see them. 15 deg is a conventional tactical-rocket limit; it was not taken from a specific source in this session, so treat it as a guess.
traj.dive_entry	trajectory	MODELLING CHOICE: <code>SPEC.md</code> section 2 does not define the coast-to-dive transition. Rule 'max_range' (default): hold altitude until level trim can no longer be held inside the alpha limit, or until Mach falls to 1.20, then dive at <code>Requirements.gamma_terminal</code> . Rule 'range': dive as soon as the current range plus the geometric dive range reaches <code>Requirements.range_min</code> . Rule 'terminal_boost': dive the moment the sustain phase ends, with no level coast at all, which keeps the most kinetic and potential energy for the endgame.

Registered name	Module	Why it is not a measured value
traj.gravity	trajectory	Constant gravity $g = 9.80665 \text{ m/s}^2$ (standard gravity, CGPM 1901). MODELLING CHOICE: the inverse-square variation is neglected, which is under 0.4 % over the 0 to 12 km band of this mission and keeps the analytic validation cases in tests/test_trajectory.py exact.
traj.guidance_gains	trajectory	MODELLING CHOICE, no source: the altitude-hold and flight-path-angle autopilot gains (0.15 1/s outer, 2.0 1/s inner), the 200 m/s commanded climb rate cap, the 45 deg commanded climb angle cap and the 20 g commanded load factor cap are arbitrary. Guidance law design is an explicit non-goal (SPEC.md section 8); these gains only have to fly the SPEC.md section 2 profile smoothly. They do not appear in any reported performance number other than through the flown trajectory. The load factor cap matters for one thing only: it keeps the recorded CN_required history finite at the dive-entry command step, so the sizing loop can read it.
traj.terminal_boost_ignition	trajectory	MODELLING CHOICE: SPEC.md section 2 has no terminal-boost phase, so the ignition trigger is invented here and exposed rather than hidden. Rule 'carry_to_impact' (default) lights the pulse when the altitude falls to $\text{terminal_ignition_margin} * V * \sin(\gamma_{\text{terminal}}) * t_{\text{terminal}}$, that is when the predicted remaining descent time equals the terminal burn time, so the pulse burns out at impact instead of above the ground. Rule 'dive_entry' lights it at dive entry. Rule 'never' leaves it unlit, which is what the two-phase regression case uses. The margin defaults to 1.0 and is a free parameter for the sizing loop: the prediction uses the CURRENT speed and the pulse accelerates the rocket, so a margin near 1.0 lights slightly late and the pulse is still burning at impact. That is the safe side; a margin above about 1.3 wastes propellant above the ground.
traj_iv1.launch_speed	trajectory_iv1	GUESS: the vertical rise starts at 1.0 m/s rather than at rest. A 3-DOF point mass at exactly zero speed has an undefined flight-path angle and a degenerate γ_{dot} equation, so the speed must be non-zero. No canister exit-velocity figure was sourced. The choice is close to harmless: at the IV-1 launch thrust-to-weight the vehicle passes 1 m/s within about 10 ms of ignition, so the programme is shifted by that much and the burnout speed changes by under 1 m/s. Exposed as the AscentMission(v_start=...) argument so it can be traded.
traj_iv1.mass_programme	trajectory_iv1	The mass programme is owned by the integrator, not by the mass statement: $m(t) = m_0 - \text{propellant burned} - \text{jettisoned mass}$. 'mass_stack' supplies only m_0 , read from a float, from 'total_mass' (config.MassStatement), from 'total' (masses.MassBuildup), from 'm0', from 'launch_mass', from a zero-argument callable, or from mass_at(0, 0, False). Nothing is guessed and no default launch mass exists: an unrecognised object raises.

Registered name	Module	Why it is not a measured value
traj_iv1.pitch_programme	trajectory_iv1	MODELLING CHOICE, and SPEC_IV1.md section 8 requires it to be declared. The vehicle rises vertically until dv.t_pitch, then turns at the constant commanded rate dv.pitch_rate_max until the flown flight-path angle reaches dv.gamma_pitch, then flies a pure gravity turn (zero commanded normal force) for the rest of the flight. There is NO autopilot gain: the commanded rate is a design variable applied open loop, and the end of the turn is found by bisecting the step that crosses gamma_pitch. There is no thrust-vector control and no side thruster, so the turn is flown on aerodynamic normal force alone and is frequently limited by the alpha limit; the limited steps are flagged rather than flown as commanded.
traj_iv1.q_floor	trajectory_iv1	MODELLING CHOICE, carried over from rocketgen.sizing.trajectory: below 200 Pa of dynamic pressure the required-normal-force coefficient is frozen at zero instead of being divided by a vanishing $q \cdot S_{ref}$. Reaching this on a lofted arc is recorded in TrajectoryResult.message and in diagnostics['q_floor_hit'].
traj_iv1.velocity_floor	trajectory_iv1	GUESS: the run is declared stalled below 20 m/s. Nothing in SPEC_IV1.md sets a floor. It exists because the gamma_dot equation divides by speed, so an arbitrarily slow point mass is numerically meaningless, and because the aero model is not valid there either. The check is ARMED only after the vehicle has once exceeded the floor, so the 1 m/s launch condition cannot trip it. Exposed as AscentMission(velocity_floor=...).

Table 22. All 55 registered values that are a guess, a modelling choice, an approximation or an assumption. No text is truncated: clipping a source string mid-word would defeat the purpose of publishing it. Source: runs/IV-1/figures/evidence_iv1.json.

7. How to reproduce this

Every artefact in this report comes from the commands below. The analysis writes JSON. The figure scripts read that JSON. This document reads it too, so no number is typed in twice.

```
.venv/Scripts/python.exe -m pytest tests -q
.venv/Scripts/python.exe scripts/iv1_converge.py --ntop
.venv/Scripts/python.exe -m rocketgen.report.evidence_iv1
.venv/Scripts/python.exe -m rocketgen.report.fig_mass_iv1
.venv/Scripts/python.exe -m rocketgen.report.fig_margins_iv1
.venv/Scripts/python.exe -m rocketgen.report.fig_infeasible_iv1
.venv/Scripts/python.exe -m rocketgen.report.fig_ascent_iv1
.venv/Scripts/python.exe -m rocketgen.report.fig_envelope_iv1
.venv/Scripts/python.exe -m rocketgen.report.build_report_iv1
```

The evidence collector does not call nTop and does not repeat the pitchover search. It rebuilds the converged design vector, checks it field by field against the recorded one, reads the stored nTop measurements back off disk, and re-flies the one converged trajectory. It then checks the launch mass, the intercept and the peak dynamic pressure against the recorded values before it writes anything. A re-flight that did not reproduce the recorded result fails there rather than publishing a second answer.

Component	Version or value
Python	3.11.15
numpy	1.26.4
scipy	1.13.1
SUAVE	2.5.2, vendored from github.com/suavecode/SUAVE
nTop Automate	5.53.2 installed, 5.54.0 development build
Notebook blocks	422
Notebook inputs	27
Block universe signatures	853
Mass-properties relative error	0.0020
Surface-area relative error	0.010
Mesh tolerance, STL export only	5.0 mm
Drag calibration applied to IV-1	no

Table 23. Environment. numpy and scipy are pinned below their current major versions, because SUAVE 2.5.2 does not run on either.

One warning about the toolchain is worth repeating. The nTop command-line tool returns exit code 72 when a block fails, not on success. Published guidance states the opposite. A notebook given an invalid dimension returns 72 and writes no output at all. Success must therefore be judged on the expected files existing and being non-empty, never on the return code alone.